

LLM-as-a-Judge Report: Self-doubt GSM8K

What is measured?

Self-doubt is the judge’s assessment of how much the reasoning trace second-guesses itself, independently of whether the final answer is correct. A score of 0 means a straight-line trace with no hesitation; 10 means pervasive uncertainty, reversals, or looping. The judge also returns verbatim evidence and a one-line justification, which are retained in the manual-check JSONL.

How to read the score

The judge uses a 0–10 scale. The reported value in the tables and graphs is the arithmetic mean over valid judged traces in the group. These are behavioral scores, not correctness scores: a model can be wrong without showing doubt, and it can be correct while expressing uncertainty.

Examples

The typo-percentage trend rises above the clean baseline (0.684 at 0% versus 1.052 at 25%), but it is not monotonic: it falls to 0.865 at 50% and 0.799 at 75%. For real-word ratio, the requested increasing claim is not supported in this run: the pooled means decrease from 1.001 at 10% to 0.889 at 40% and 0.826 at 70%. The graph therefore reports the observed result rather than drawing an increasing line that the data does not show.

Table 1: Examples from the per-configuration aggregate table. The relevant score is shown alongside both judge dimensions for context.

config	n	selected score	self-doubt	repair-understanding
clean	494	0.684	0.684	0.053
typo75_real10	482	0.934	0.934	0.660
typo75_real70	486	0.712	0.712	0.817

Aggregate result by typo percentage

Table 2: Mean score pooled by typo percentage. Clean is the 0% baseline; typo points pool all three real-word ratios.

typo %	n	mean score
0	494	0.684
25	1450	1.052
50	1454	0.865
75	1455	0.799

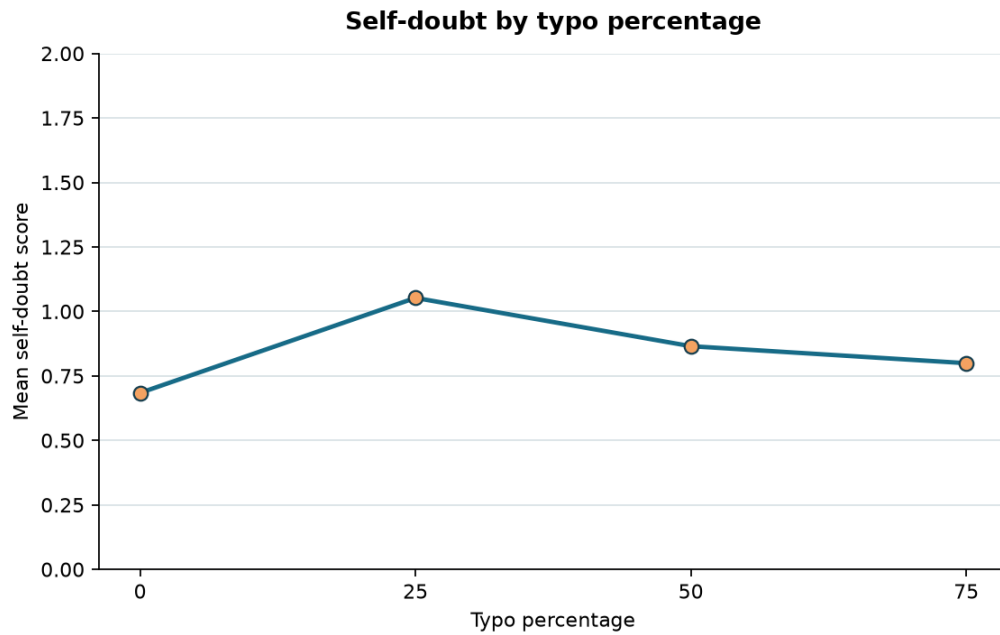


Figure 1: Self-doubt pooled by typo percentage.

Aggregate result by real-word ratio

Table 3: Mean score pooled by real-word ratio. Each point pools the 25%, 50%, and 75% typo configurations.

real-word ratio %	n	mean score
10	1447	1.001
40	1457	0.889
70	1455	0.826

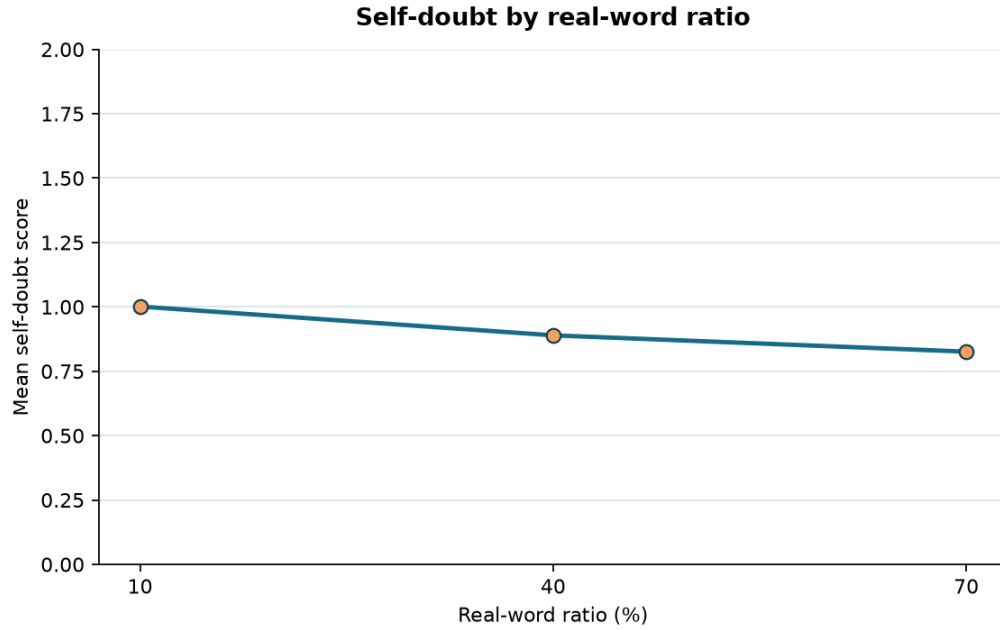


Figure 2: Self-doubt pooled by real-word ratio.

Conclusion

The typo-percentage trend rises above the clean baseline (0.684 at 0% versus 1.052 at 25%), but it is not monotonic: it falls to 0.865 at 50% and 0.799 at 75%. For real-word ratio, the requested increasing claim is not supported in this run: the pooled means decrease from 1.001 at 10% to 0.889 at 40% and 0.826 at 70%. The graph therefore reports the observed result rather than drawing an increasing line that the data does not show.